

Evaluation Problem 1

Group C21

Duration: 2 hrs

Instruction:

Do not use help from internet. Your code will be checked for plagiarism. Do not share your code with friends, in case of plagiarism both parties will get 0.

Solve both problems.

1. Implement a parent process that monitors the "health" of a child process. The child process performs a long-running task. Every 1 second, the child must send a message through a pipe to the parent. The parent process sets an alarm(3). If the parent receives the message from the pipe, it resets the alarm. If the SIGALRM handler is triggered (meaning no message was received for 3 seconds), the parent assumes the child is "hung." The parent then uses kill(child_pid, SIGKILL) to terminate the child, cleans up with wait(), and spawns a fresh child process to restart the task. [10]
2. You have a massive text file (or a directory of 20 small files) and a target keyword. The main thread determines the number of lines or files. It spawns N threads (e.g., 4 threads). Each thread is responsible for searching a specific segment (e.g., Thread 1 searches lines 1–100, Thread 2 searches 101–200). Each thread should count how many times the keyword appeared in its segment. Threads must return their individual counts to the main thread using pthread_exit() or by updating a shared structure. The main thread must use pthread_join() to collect all results and print the total count. [10]